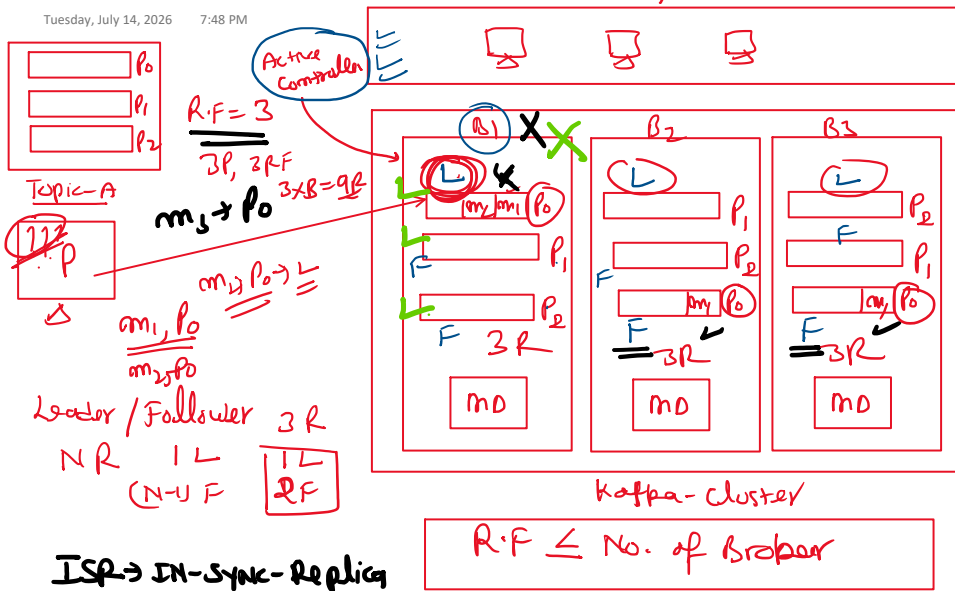
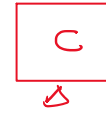


Zookeeper / Kraft



Kafka

- No msg Log
- FT
- HA
- 2FT
- R.F.



P - Client

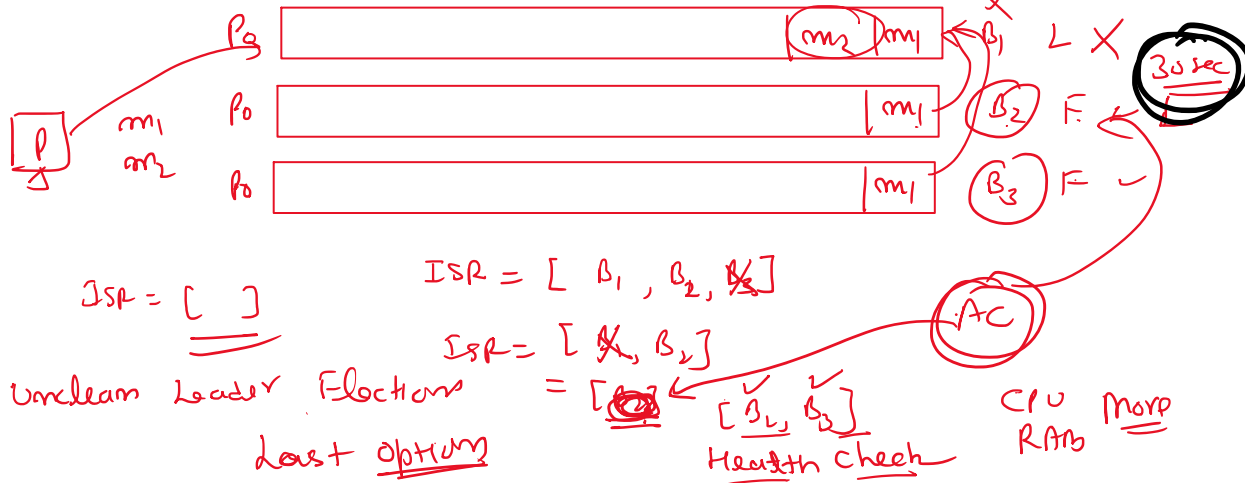
C - Client

P - Write

C - Read

P - XW

C - XR



ISR = []

ISR = [B1, B2, B3]

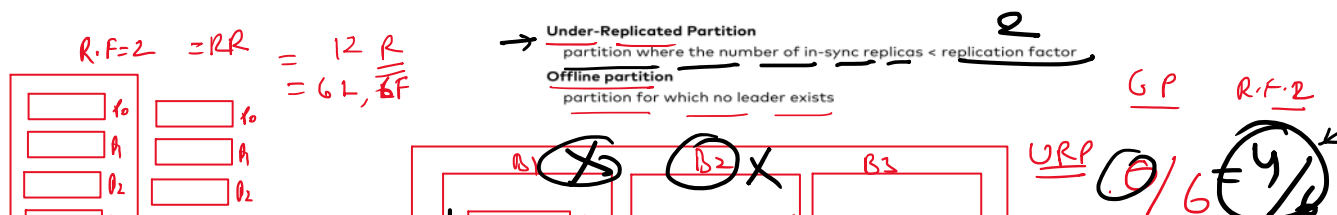
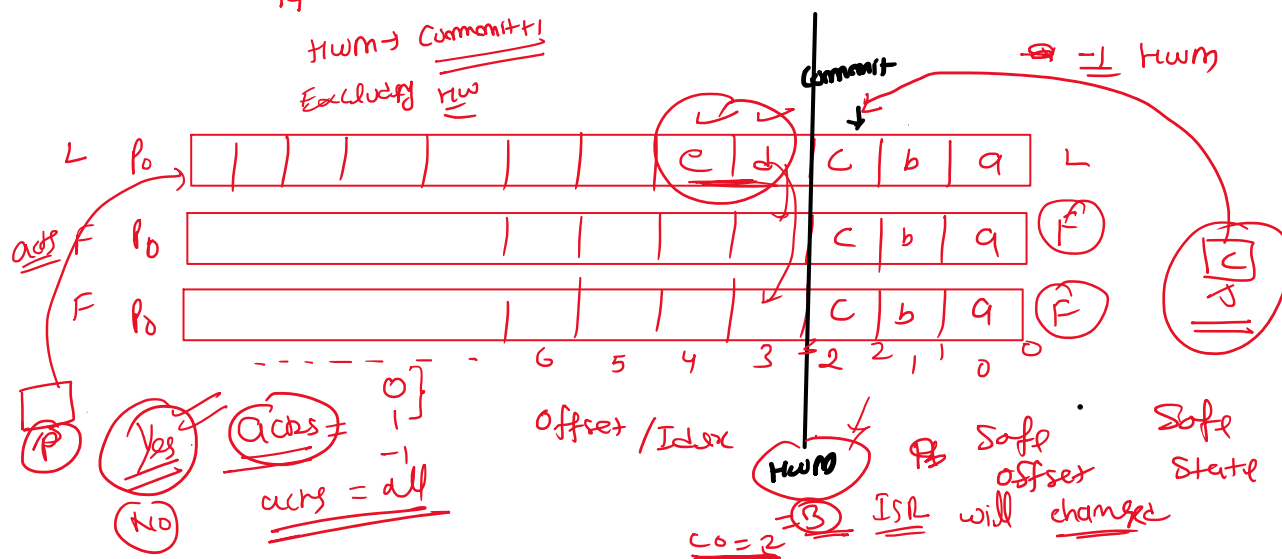
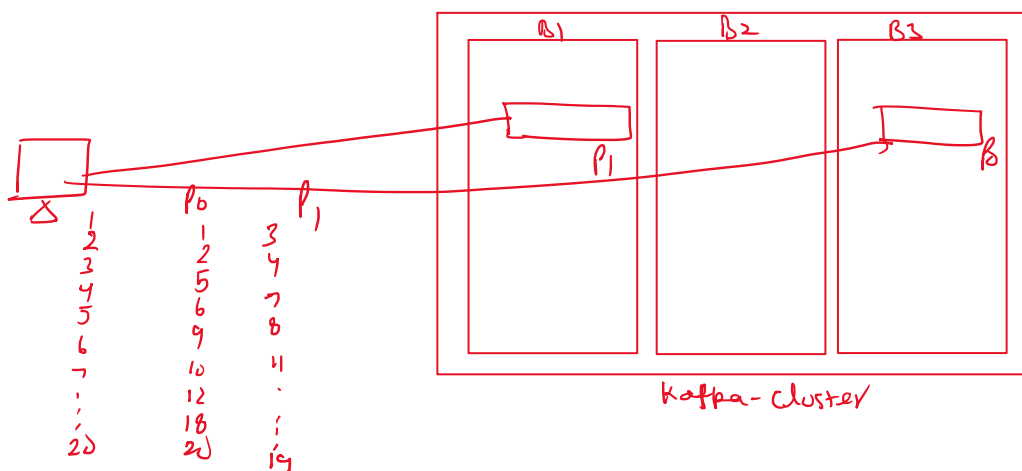
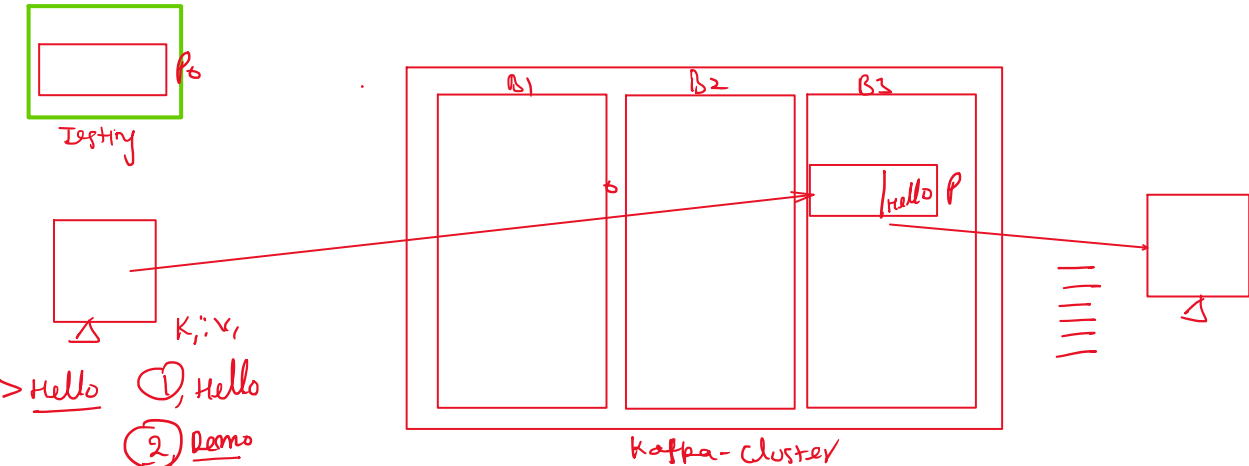
ISR = [B1, B2]

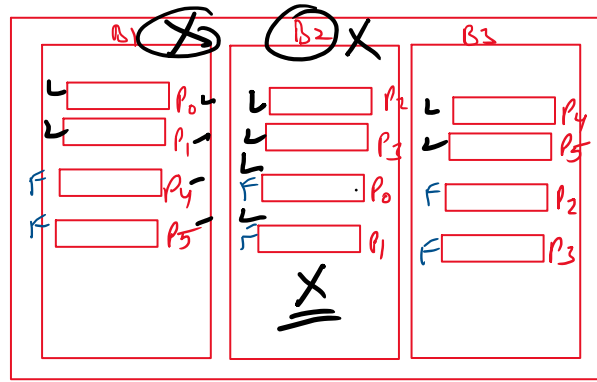
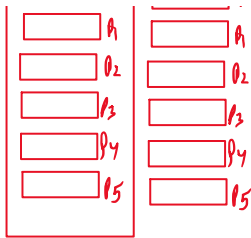
Unclean Leader Election

Least option

Health check

- Producer/Consumer
- Kafka Cluster
- Broker
- Topic
- A/C CMD
- Partition
- Replication factors
- Leader/follower
- Leader Election
- ISR
- Unclean Leader Election

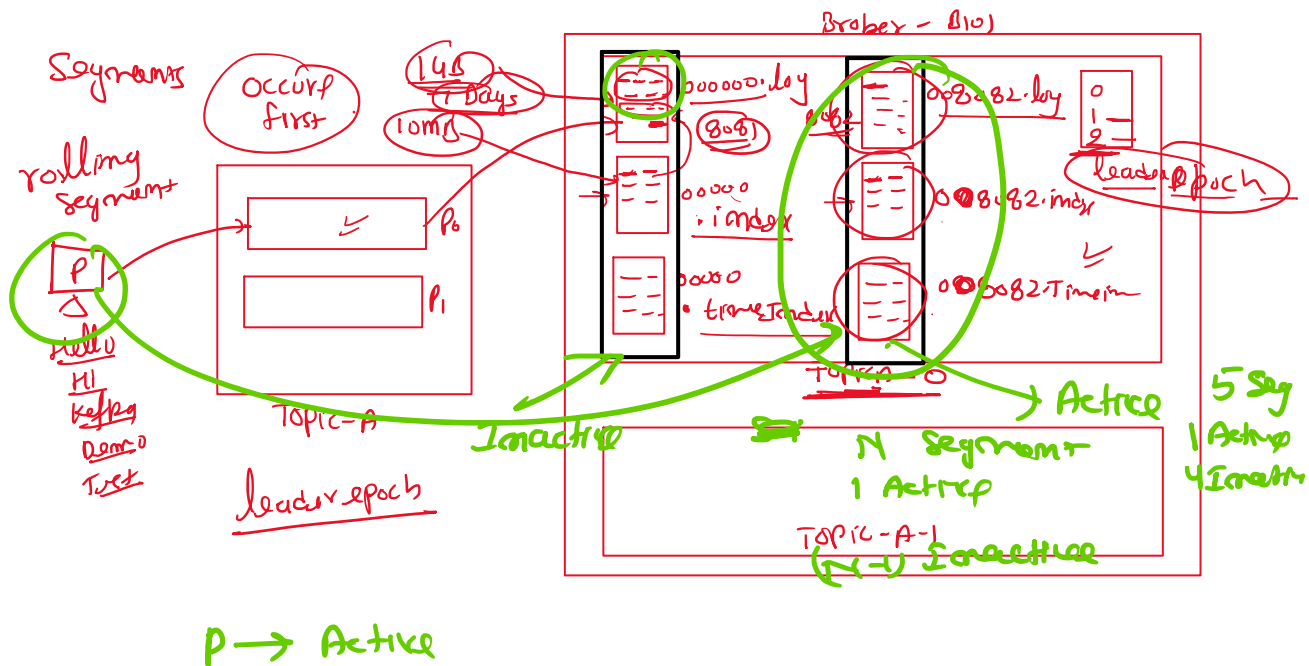




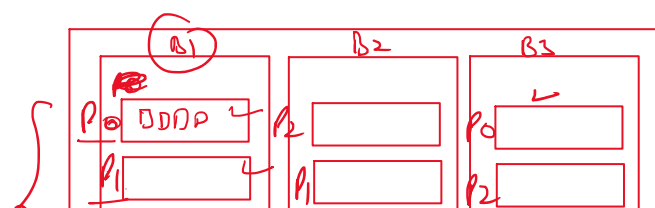
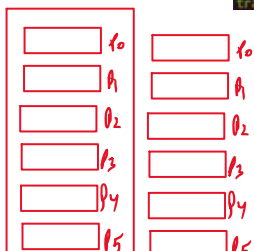
Kafka-cluster

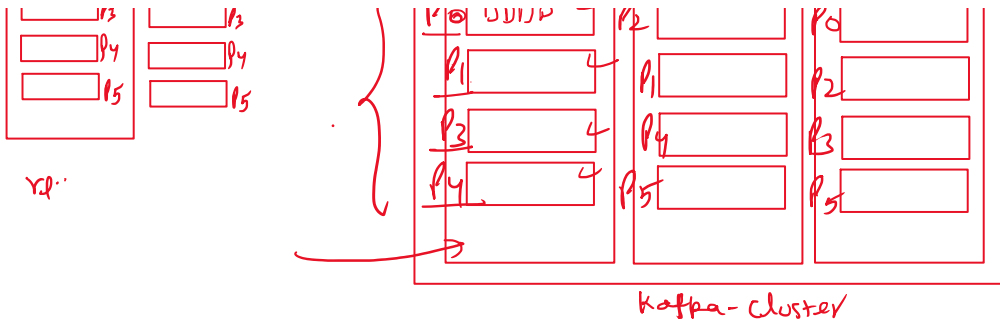
URP $\frac{0}{6} = \frac{4}{6}$
 OP $\frac{0}{6} = \frac{2}{6}$
 High Risk

LAB 1 TIME: 20 MINS
 LUNCH TIME: 40 MINS
 REUME BACK IN 80 MINS



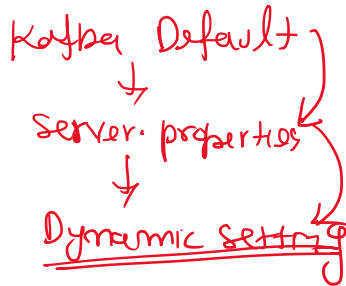
```
Topic: replicated-topic TopicId: 8fMxMnp9Ti6UQWfthNrg PartitionCount: 6 ReplicationFactor: 2 Configs:
Topic: replicated-topic Partition: 0 Leader: 3 Replicas: 3,1 Isr: 3,1 Offline: Elr: N/A LastKnownElr: N/A
Topic: replicated-topic Partition: 1 Leader: 1 Replicas: 1,2 Isr: 1,2 Offline: Elr: N/A LastKnownElr: N/A
Topic: replicated-topic Partition: 2 Leader: 2 Replicas: 2,3 Isr: 2,3 Offline: Elr: N/A LastKnownElr: N/A
Topic: replicated-topic Partition: 3 Leader: 3 Replicas: 3,1 Isr: 3,1 Offline: Elr: N/A LastKnownElr: N/A
Topic: replicated-topic Partition: 4 Leader: 1 Replicas: 1,2 Isr: 1,2 Offline: Elr: N/A LastKnownElr: N/A
Topic: replicated-topic Partition: 5 Leader: 2 Replicas: 2,3 Isr: 2,3 Offline: Elr: N/A LastKnownElr: N/A
training@training:~/confluent-admin$
```





CH-1
CH-4

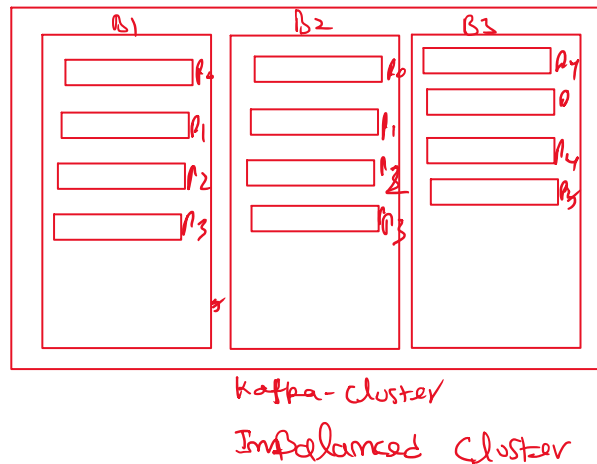
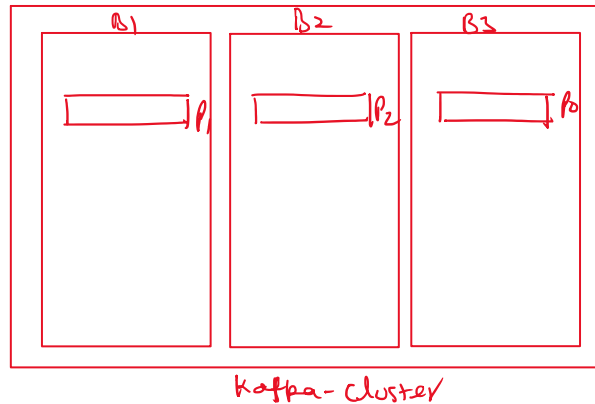
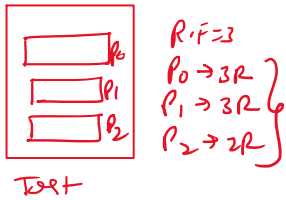
- Kafka Easy Fundamentally
- Produce msg Rel.
- Partition are placed on Broker



files are created physically

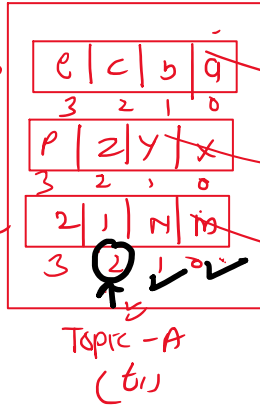
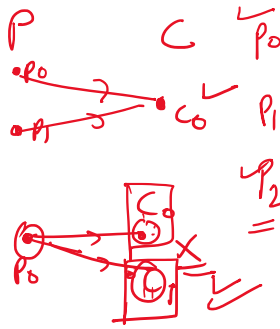
- log
- index
- time Index

Topic: test	TopicId: 6Ho9KH0TRnms8r-cFruUDQ	PartitionCount: 3	ReplicationFactor: 1	Configs:		
Topic: test	Partition: 0	Leader: 3	Replicas: 3	Isr: 3	Offline: Elr: N/A	LastKnownElr: N/A
Topic: test	Partition: 1	Leader: 1	Replicas: 1	Isr: 1	Offline: Elr: N/A	LastKnownElr: N/A
Topic: test	Partition: 2	Leader: 2	Replicas: 2	Isr: 2	Offline: Elr: N/A	LastKnownElr: N/A

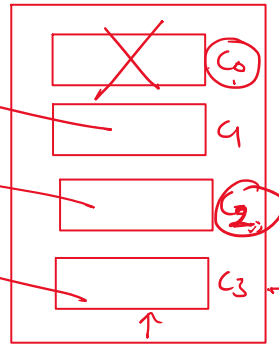


CCC
Res flay

$P \rightarrow C$



C_0, C_1, C_2, C_3



$4 > 3$
 $C > P$
 $C - P = \underline{\text{Idle}}$
 $4 - 3 = 1$

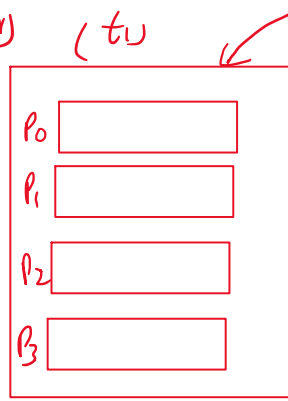
-- Consumer-offset

Event (Rebalancing Trigger)

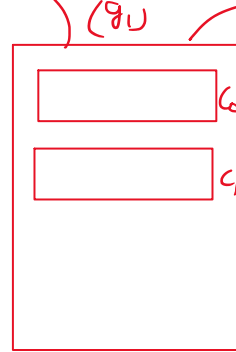
$C < P$

- 1- Added Consumer
- 2- Deleted "
- 3- Partition Increase
- 4- Subscription Change

Rebalancing

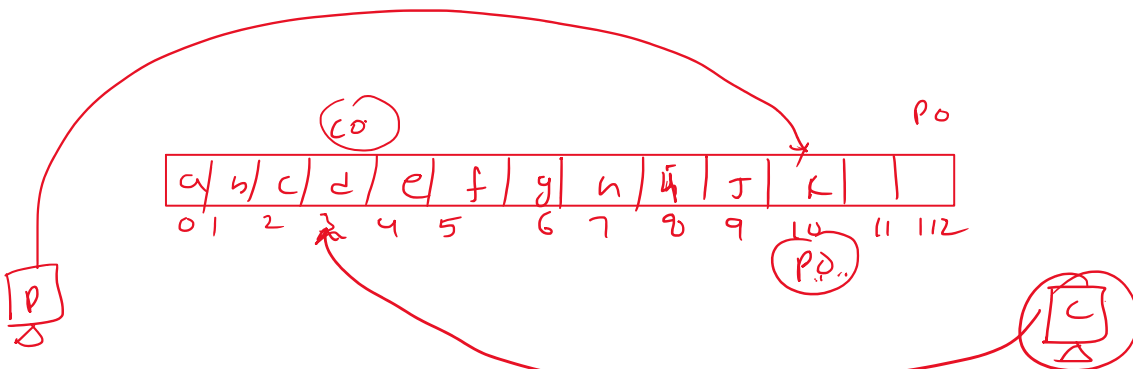


Subscribe



Avoid RB

Consumption Pause



$$P_0 - C_0 = \underline{\text{lag}}$$

$$10 - 3 = \underline{7}$$

$$\text{lag} = \underline{7}$$

$$\neq 0$$

